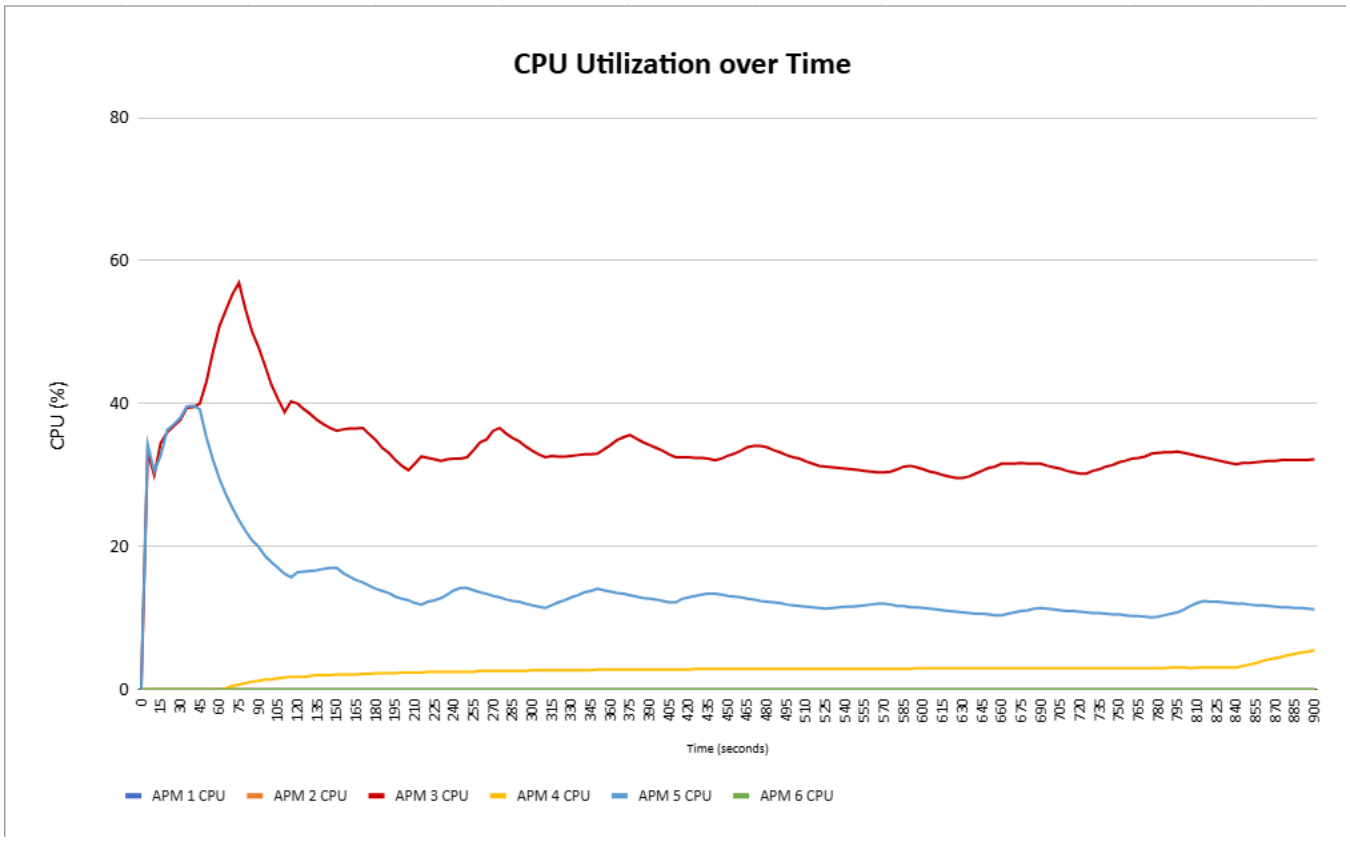
NSSA-220 Project 1: Application Performance Monitoring

Elvis Lin, Anuj Agrawal, Jeliel Brown

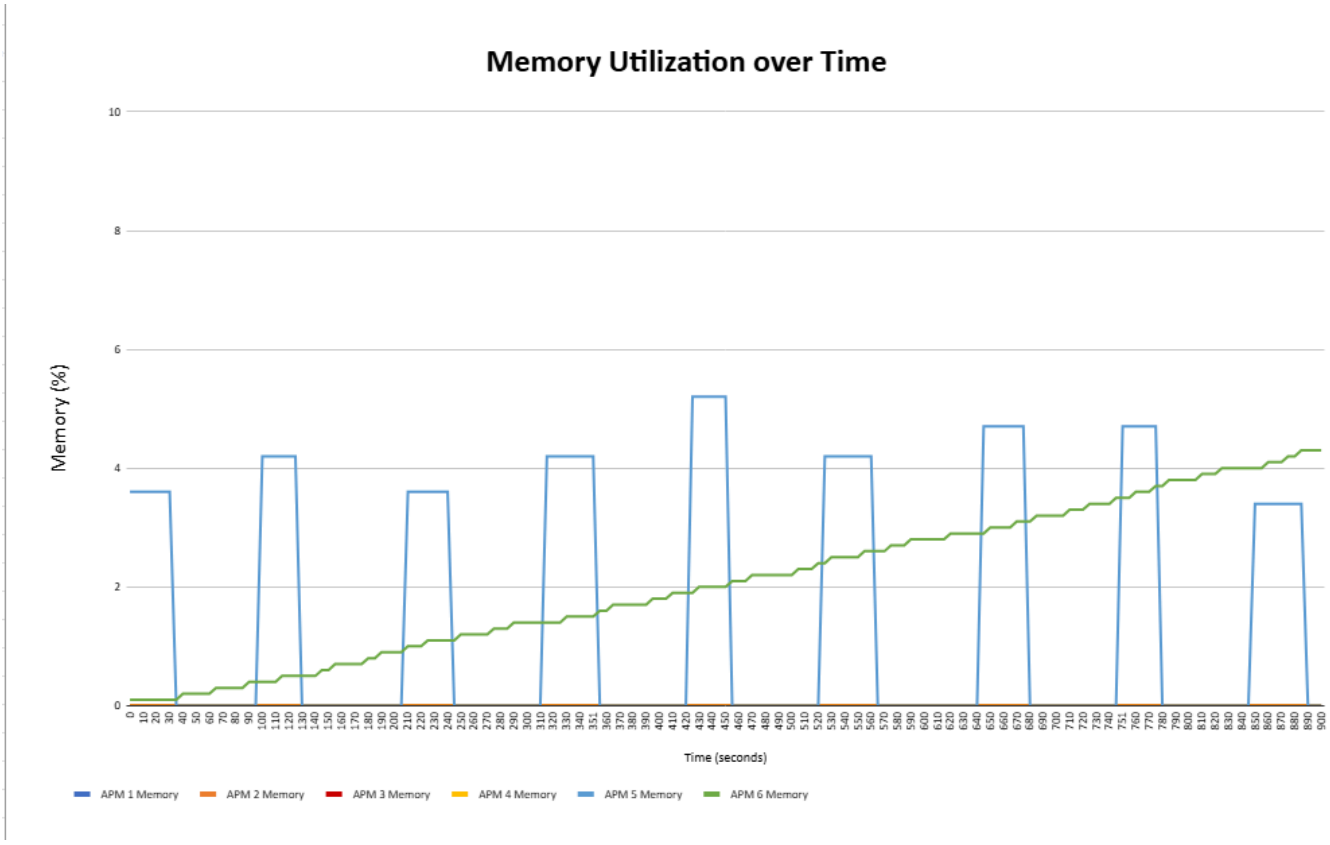
Introduction

This project focused on building an Application Performance Monitoring (APM) tool using a Bash script. The goal was to run multiple applications and continuously monitor their performance by collecting process-level and system-level metrics. These metrics included CPU usage, memory usage, network activity, and disk performance, which were recorded over time in CSV files. The collected data was then analyzed using graphs to better understand system behavior and resource utilization.

Process Level Metrics



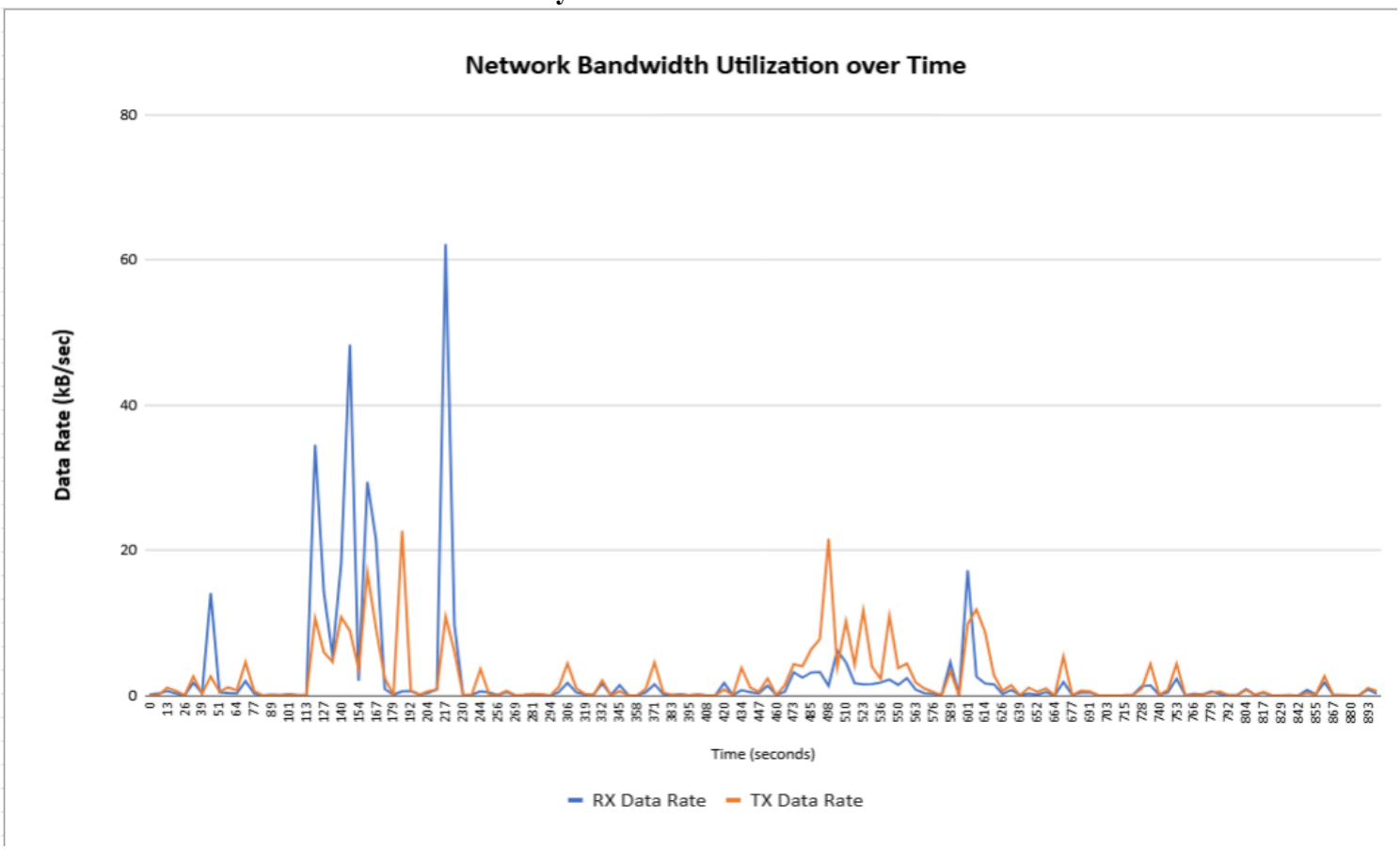
The plot shows CPU utilization for six APMs over time, with APM 3 (the red line) consistently using the most CPU, stabilizing around 30-40% with a peak at around 56.9 %. APM 5 (blue) starts high with a peak at 39.7%, then quickly drops and fluctuates in the mid-teens to around 15-20%, while APM 4 (yellow) slowly increases from near 0 up to about 6-7%. The other APMs (1, 2, and 6) use almost no CPU throughout the period. The CPU utilization plot shows how much processing power each application used over time. The values fluctuate, indicating periods where the application was more or less active.



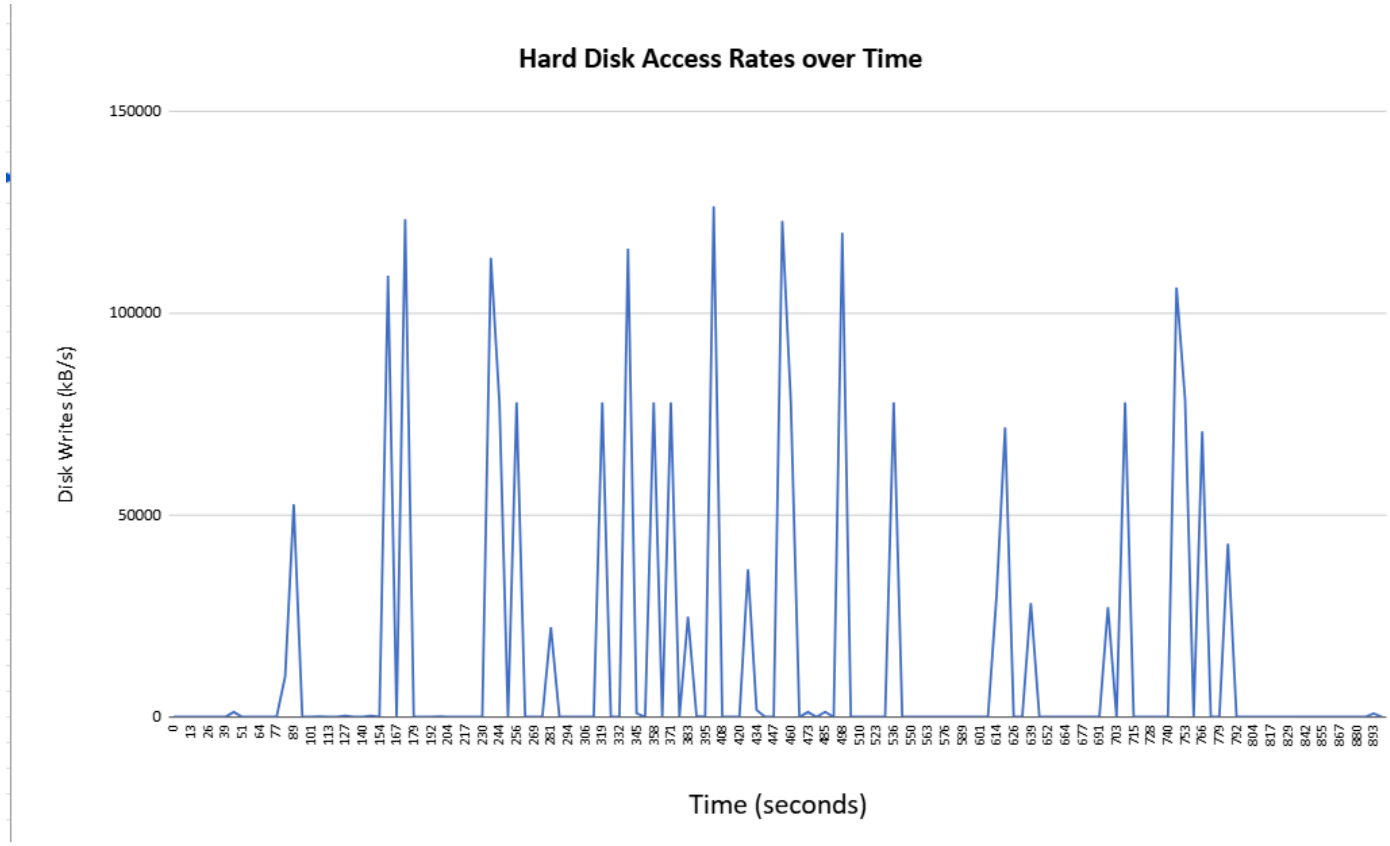
The plot shows memory utilization for six APMs over time, but only APM 5 and APM 6 actually consume noticeable memory. APM 6's usage (the green line) increases gradually and steadily from near zero to about 4.3 units, while APM 5's usage the (blue line) jumps to high levels intermittently and then returns to zero between spikes.

APM 3 used the most sustained CPU, with APM 5 also relatively high but then drops and fluctuates around 15-20 percent, while APMs 1, 2, and 6 used essentially no CPU, and APM 4 stayed low but gradually increased. For memory, only APM 5 and APM 6 used notable amounts: APM 5 showed sharp, periodic spikes that returned to zero, whereas APM 6 increased steadily over time. That steadily rising APM 6 memory line, without corresponding drops, looks like a potential memory leak.

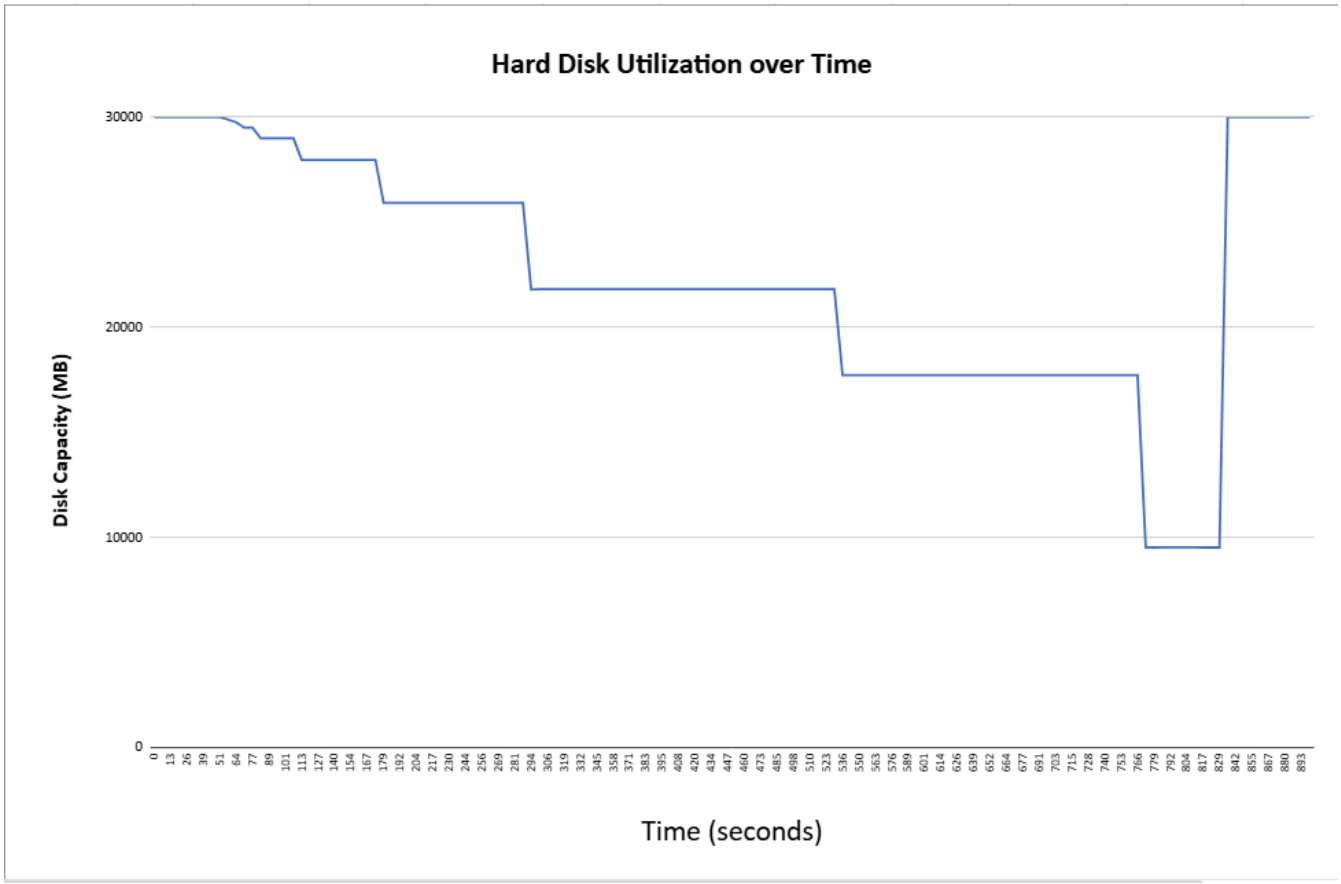
System Level Metrics



Network bandwidth utilization is bursty rather than steady. The receive rate is generally higher and more variable than the transmit rate, with several sharp spikes early with RX jumping to about 60 kB/sec, in the timeline and a smaller cluster of activity later, with both rates staying near zero for much of the capture.



The hard disk access rate is highly bursty over time, with long stretches of little to no disk writing interrupted by sharp spikes. These peaks show that disk activity happens in short, intense bursts rather than at a steady, continuous rate.



The hard disk utilization plot shows available disk space steadily decreasing in large steps from about 30,000 units down to around 10,000 over time, indicating progressive consumption of storage. Near the end, disk space jumps sharply back up to about 30,000 units, suggesting a cleanup or deletion of large amounts of data

The transmit and receive data rates were somewhat similar in overall pattern, since both stayed near zero to five kB/ sec for most of the run and only rose during short bursts of activity. However, the receive rate was more variable and reached higher peaks than the transmit rate, especially early in the timeline, which suggests incoming traffic was heavier and less consistent than outgoing traffic. The hard disk access rate showed a very different pattern, with repeated sharp spikes separated by long idle periods, indicating bursty write activity rather than steady use. Hard disk utilization, meanwhile, appeared to decline in steps and then jump back up near the end, which is consistent with storage gradually being used and then a large cleanup or deletion occurring.

Summary and Lessons Learned

Overall, the VM appeared to have sufficient computing resources for the workload: CPU usage stayed well below saturation, memory pressure was moderate aside from one process with a possible leak, network traffic was mostly idle with only brief spikes, and disk space, while steadily consumed, never actually ran out before a cleanup. The main concern was the steadily increasing memory usage of one APM, which suggests we'd want better monitoring and limits in

a production setting. As a team, we learned the importance of dividing responsibilities clearly (data collection, analysis, visualization), communicating frequently about assumptions, and using shared tools/documentation so everyone interpreted the performance metrics consistently.